MIL-P-81305(AS)
Amendment 4
19 March 1976
SUPERSEDING
Amendment 3
7 December 1971

MILITARY SPECIFICATION

POWER DISTRIBUTION PANEL, ALTERNATING CURRENT,
F-4 AIRCRAFT, SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-P-81305(WP), 17 August 1965 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-P-81305(WP) is changed to MIL-P-81305(AS) and throughout the specification Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 1, paragraph 1.1, Revise to read: "This specification covers the requirements for power distribution panels designed for use in 30KVA and 40KVA, 115/200 volt, 400 Hz, 4 wire grounded neutral, wye connected AC electric power systems on F-4 aircraft. Both panels are designed for use in conjunction with the generating system components shown on MS90302. Relays and contactor coils in this panel are energized from a nominal 28 volt DC source."

Add "1.1.1 Military Part Numbers 30KVA Panel P/N M81305-1
40KVA Panel P/N M81305-2"

Page 2, paragraph 3.2 Dimensions - Revise to read: "The dimensions of the panels shall conform to Figure 1 for M81305-1 panels and to Figure 4 for M81305-2 panels."

Page 3, figure 1, add to notes: "There shall be 0.41 + .010 - .000 clearance around the center hole in each of the four mounting feet."

Page 5, paragraph 3.5.1 Housing: Revise to read: "The panel housing shall be constructed of fiber glass or equivalent nonmetallic material."

Page 5, insert paragraph "3.5.2.2 Seat - Terminal seats as shown on Figure 1 shall be integral with the terminal stud."

Page 6, paragraph 3.8 Relays - Revise the first two sentences to read: "All relays and contactors used in this panel shall be hermetically sealed and shall meet the requirements of MIL-R-6106 except that the minimum number of operating cycles shall be 50,000 in lieu of 100,000. They shall be tested and meet the requirements of sampling plans A and B of MIL-R-6106 and meet the rating and operating characteristics specified in paragraph 3.8.1 and paragraph 3.8.2 of this specification."

FSC 6110

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Page 6, paragraph 3.8 Relays - Delete the last sentence.

Page 6, paragraph 3.8.2 Operating Characteristics - Add: "The relays and contactors shall be designed for operation from a nominal 28 volt DC system and shall have the following operating characteristics.

Page 7, paragraph 3.9 Current Transformers - Revise sub-paragraph (f) to read: "(f) Accuracy @ 400 Hz with sine wave current between 25 and 111 amperes, and with a line power factor of 0.75, the phase angle shall be less than 2° and the ratio error shall be less than 5%."

Page 7, paragraph 3.11 Wiring - Revise to read: "Internal wiring, and external wiring for the purpose of testing of the AC power distribution panels, shall conform to the requirements of Figure 2 for M81305-1 panels and Figure 5 for M81305-2 panels."

Page 7, paragraph 3.12 Identification of Product - Revise to read: "The panels shall be marked in accordance with MIL-STD-130 and shall include the following information.

AC Power Distribution Panel, F-4 Aircraft
Mfr's Part Number
Mfr's Name and Code Identification
Mfr's Serial Number
115/200V, 400Hz, 4 Wire
Federal Stock Number
Date and Source Code (In accordance with MIL-STD-456)"

Page 8, Figure 2: Change to the Figure 2 attached to this amendment.

Page 9, paragraph 4.3.1 Sampling Instruction - From the fifth item in the list of information to be attached to test samples. Delete M81305-1 and add: "(M81305-1 or M81305-2)."

Page 12, paragraph 4.4.3.2 Tests - Revise first two lines to read: "Sampling plan C tests shall consist of all tests on sample number 1 and sample number 3 specified in Table 1 with the following exceptions."

Page 14, paragraph 4.6.6.1 High Current (Power) Circuits - Under table-source-voltage-28 volts DC, change all "175" MV voltage drops to "325" MV change all "325" voltage drops to "430" MV.

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Page 15, paragraph 4.6.7.1 Accuracy - Revise to read: "The accuracy of the current transformers (CT) shall be measured at the frequency line power factor, sine wave current, and burden specified in paragraph 3.9. The ratio and phase angle errors shall be not greater than values specified in 3.9. When calculating the phase angle error, the error introduced in the measuring instruments, due to the ratio error shall be calculated by the following formula and shall be positive in algebraic sign:

$$\text{o/o Ratio Error} = \frac{(\text{Measured Current Ratio}) - (\text{Design Ratio})}{(\text{Design Ratio})} \times 100"$$

The contractor shall furnish quantitative test data certified by the appropriate Government Quality Assurance Representative to demonstrate compliance with accuracy requirement herein."

Page 17, paragraph 4.6.8 Simultaneous Operation on Pickup - Revise to read: "With the nominal voltage applied, all main "NO" contacts on any one relay or contactor shall operate simultaneously within 5 milliseconds and all auxiliary "NO" contacts on any one relay or contactor shall operate simultaneously within 15 milliseconds."

Page 17, paragraph 4.6.10 Functional Tests - Delete subparagraph F.

Page 18, paragraph 4.6.13 Vibration - Revise to read: "The panel shall be subjected to the vibration test procedure XII of MIL-E-5272 except that all tests shall be done at room ambient temperature only. Vibration cycling shall be run consecutively with the relays de-energized, energized at maximum pickup voltage and energized at nominal voltage, accumulating an equal number of full cycles for each condition. Vibration at resonance shall also be equally divided for each condition. During this test, all closed contacts shall remain closed and all open contacts shall remain open. A suitable indicating device shall be used to determine the ability of the relay contacts to remain in the proper position. Contact chatter in excess of 10 microseconds shall constitute failure. Upon the completion of this test, the panel shall successfully complete the functional test 4.6.10. Any structural, mechanical or electrical failure shall be cause for rejection."

Page 19, paragraph 4.6.17 Vibration Scan - Delete this paragraph in its entirety.

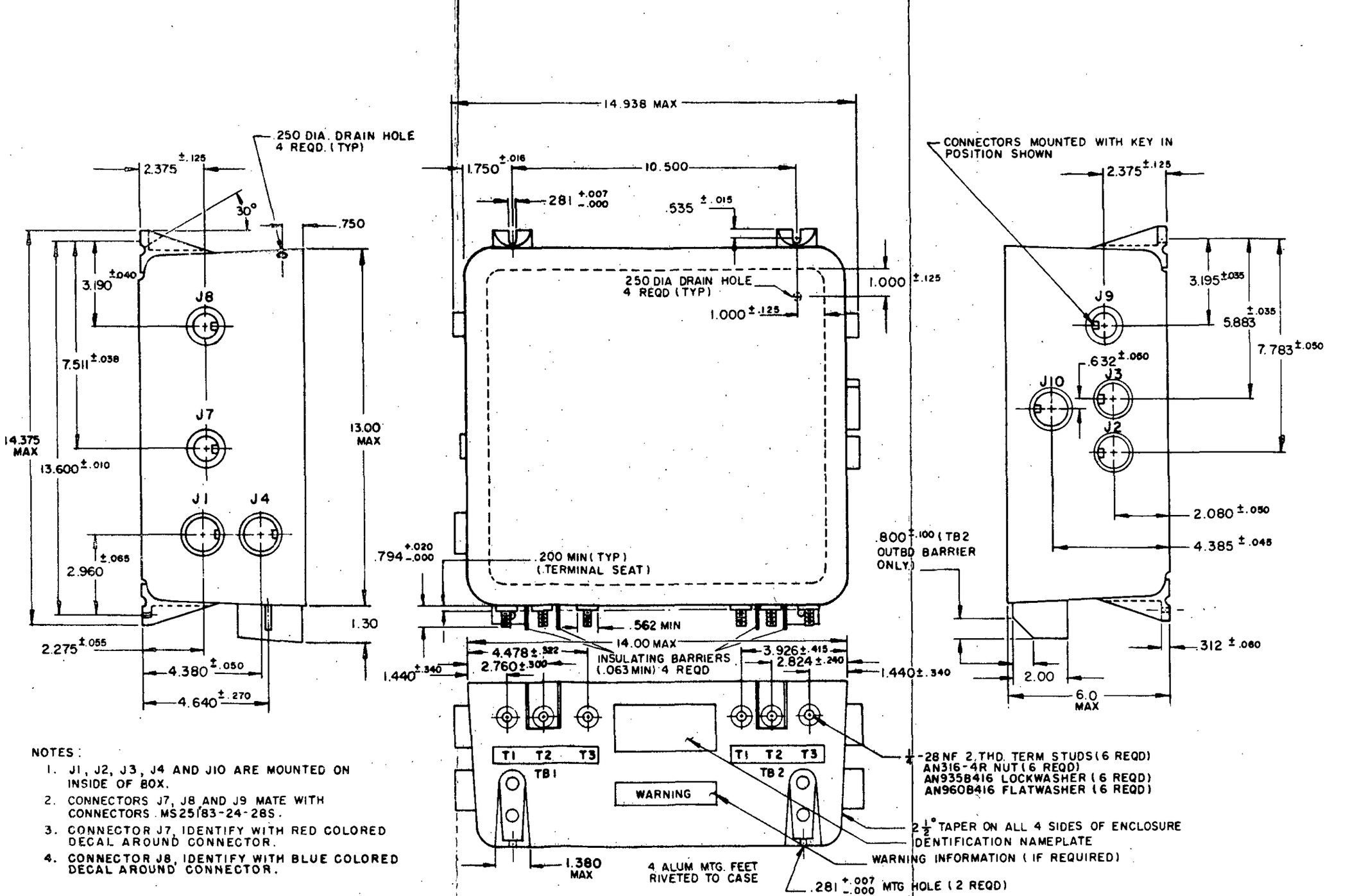
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Page 20, paragraph 6.3 - Revise last 2 sentences to read: "The activity responsible for qualification of the panels manufactured in accordance with this specification is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360. Information pertaining to qualification testing and authorization to submit products for qualification may be obtained from the Commanding Officer, Naval Air Test Center, Patuxent River, Maryland 20670, Attention: WST-33."

Preparing Activity
Navy(AS)
Project Number 6110-N198

*Denotes change from previous amendment.

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NOTES:

1. J1, J2, J3, J4 AND J10 ARE MOUNTED ON INSIDE OF BOX.
2. CONNECTORS J7, J8 AND J9 MATE WITH CONNECTORS MS25183-24-28S.
3. CONNECTOR J7, IDENTIFY WITH RED COLORED DECAL AROUND CONNECTOR.
4. CONNECTOR J8, IDENTIFY WITH BLUE COLORED DECAL AROUND CONNECTOR.

FIGURE 4
M 81305-2 OUTLINE DRAWING

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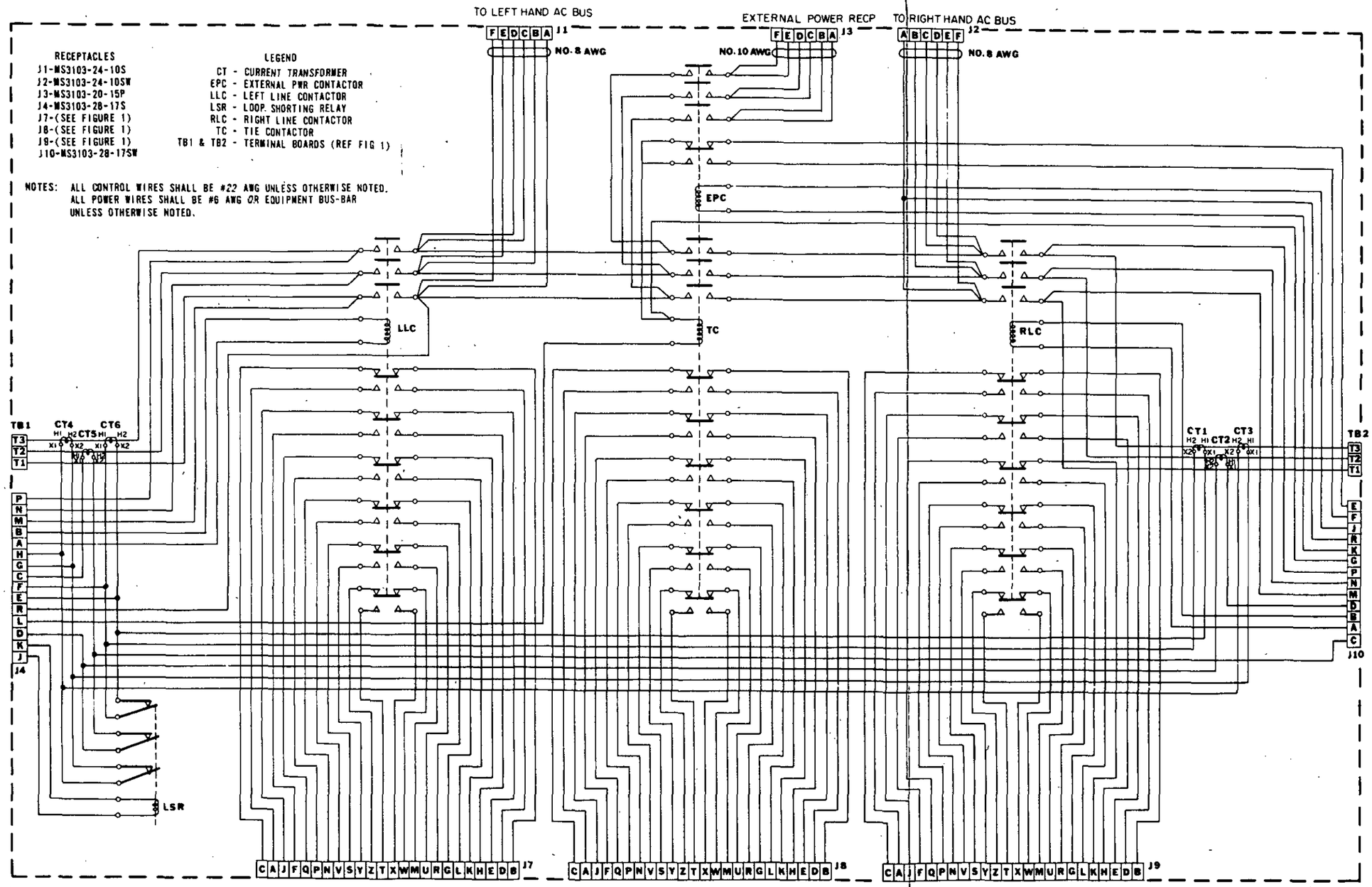


FIGURE 5
M81305-2 SCHEMATIC DRAWING

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